

Project Proposal

Name of the Project:

Anomaly-Based Accident Detection and Faulty Vehicle Identification System

The URL of the Project Web Page:

advis-ai.com.tr

Names of the Team Members:

Ahmet Faruk Sevim

Muhammet Emir Dizdar

Utku Çetin

Semih Uçan

Hasan Uzun

Project Description:

This project introduces an end-to-end intelligent traffic surveillance pipeline designed to automate accident detection and liability assessment using advanced deep learning techniques. Unlike traditional rule-based systems, this model utilizes an Anomaly Detection framework to continuously monitor traffic flow and identify collisions or irregular driving patterns in real-time. By leveraging spatio-temporal features, the system effectively filters normal traffic behavior and triggers an automated analysis whenever a high-confidence anomaly is detected, ensuring efficient use of computational resources.

Once an incident is triggered, the system extracts a precise event clip and employs a Video Classification model to categorize the accident type, such as rear-end collisions or red-light violations. Simultaneously, the pipeline performs causality analysis by mapping anomaly heatmaps to identified vehicle trajectories, allowing the system to pinpoint the primary faulty vehicle without manual intervention. The final stage integrates a high-precision License Plate Recognition (LPR) module that focuses exclusively on the liable actor to document legal evidence. This multi-layered approach transforms standard traffic cameras into autonomous diagnostic tools, providing objective, data-driven reports for insurance, legal, and law enforcement purposes.